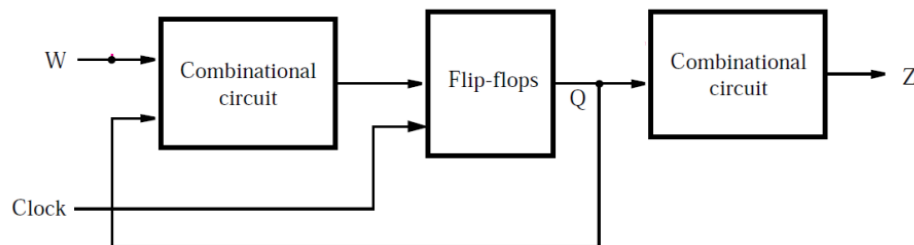
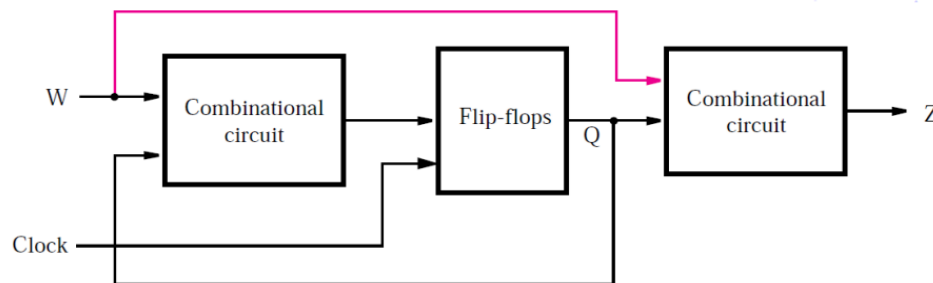


- Sequential circuits are an integral part of digital system design. Unlike combinational circuits that generate outputs solely based on the current inputs, sequential circuits incorporate memory elements that enable them to store and remember past inputs (the past behavior of the circuit).
- It is a circuit in which the values of the outputs depend not only on the present values of the inputs but also on the past behavior of the circuit. Such a circuit contains combinational circuits and storage elements (Flip flops/Latches).
- Synchronous sequential circuits commonly use flip-flops and are designed as finite-state machines (FSM).
- The name derives from the fact that the functional behavior of these circuits can be represented using a finite number of states.
- The sequential circuits (FSMs) whose outputs depend only on the state of the circuit are Moore-type:



- The sequential circuits (FSMs) whose outputs depend on both the state and the primary inputs are Mealy type.



- Counters are sequential circuits (Moore-type) that generate a sequence of binary numbers in a specific order.

Review: Design Steps

- Define the Problem and obtain the specification of the desired circuit:
 - Identify the inputs, outputs, and the desired behavior of the circuit.
- Design the **State Diagram**: represent all possible states of the system and the transitions between them. Specify the starting state.
- Create a **state table** from the state diagram.
- **State optimization** (reduction): minimize the number of states.
- **State assignment**: decide on the number of state variables needed to represent all states and perform the state assignment (binary code).
- **State assignment table**: shows binary coded states.
- Choose the type of **flip-flops**: common types include D flip-flops, JK flip-flops, or T flip-flops.
- Derive the simplified flip-flop **input equations** and **output equations** using K-map.
- Implement the circuit: draw **circuit diagram** that consists of FFs and gates.

- Example 1 (Moore-type):

Design a sequential circuit (FSM) that meets the following specifications:

1. The circuit has one input, w , and one output, z .
2. All changes in the circuit occur on the positive edge of the clock signal.
3. The output z is equal to 1 if during three immediately preceding clock cycles, the input w was equal to 1, 0, then 1. Otherwise, the value of z is equal to 0. In other words, the circuit must detect the sequence 101. So, $z=1$ if the input w was 101.

To analyze the problem let us assume the following sequence of inputs:

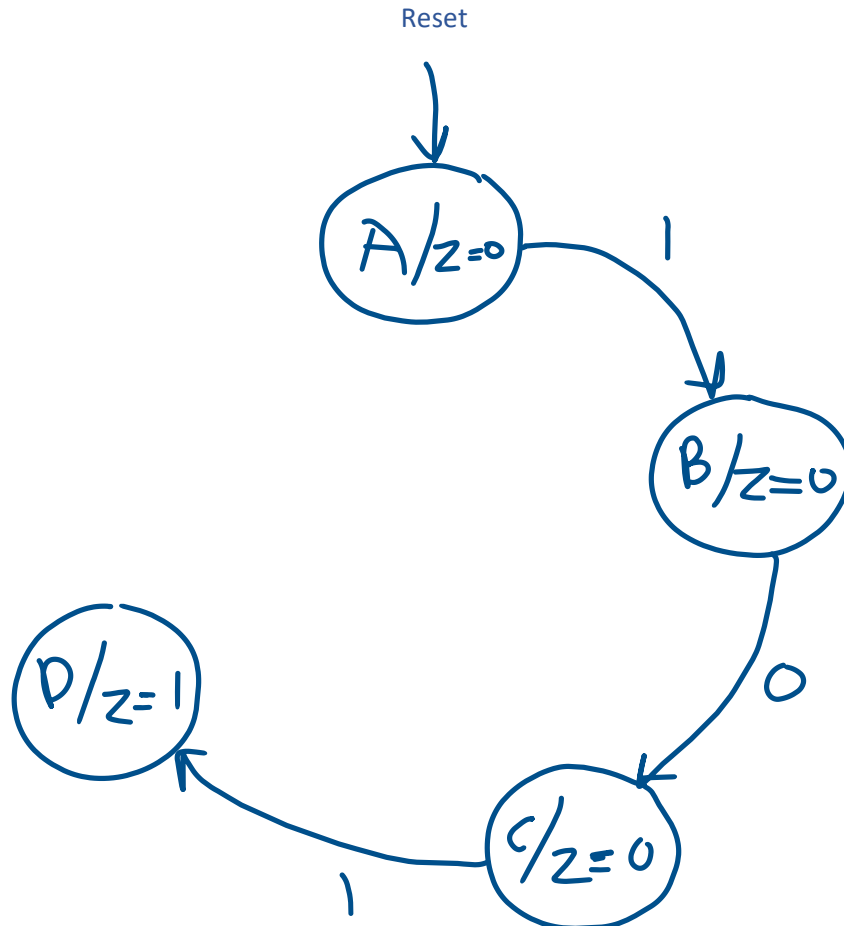
w : 0 0 1 0 1 1 0 1 0 1 0 0

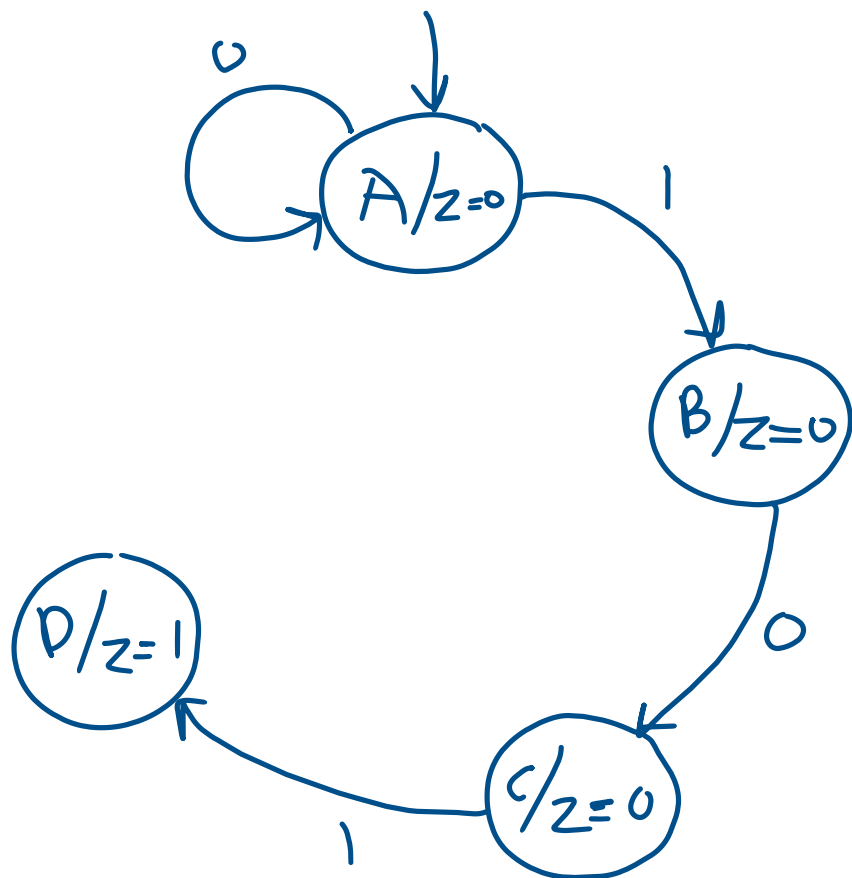
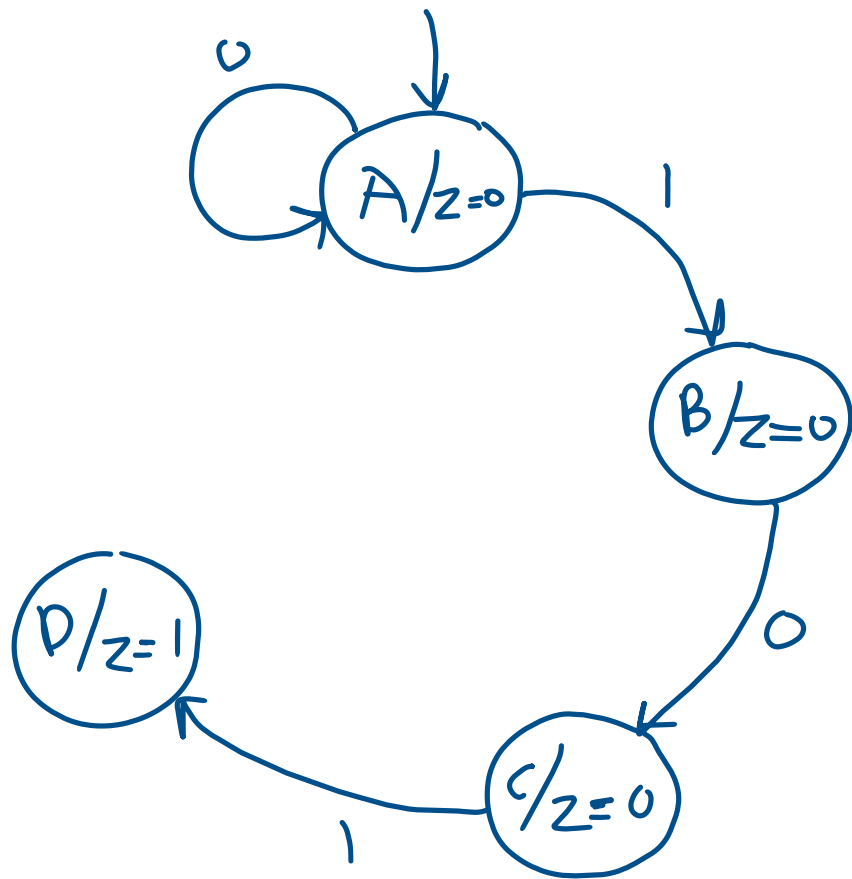
z : 0 0 0 0 0 1 0 0 1 0 1 0

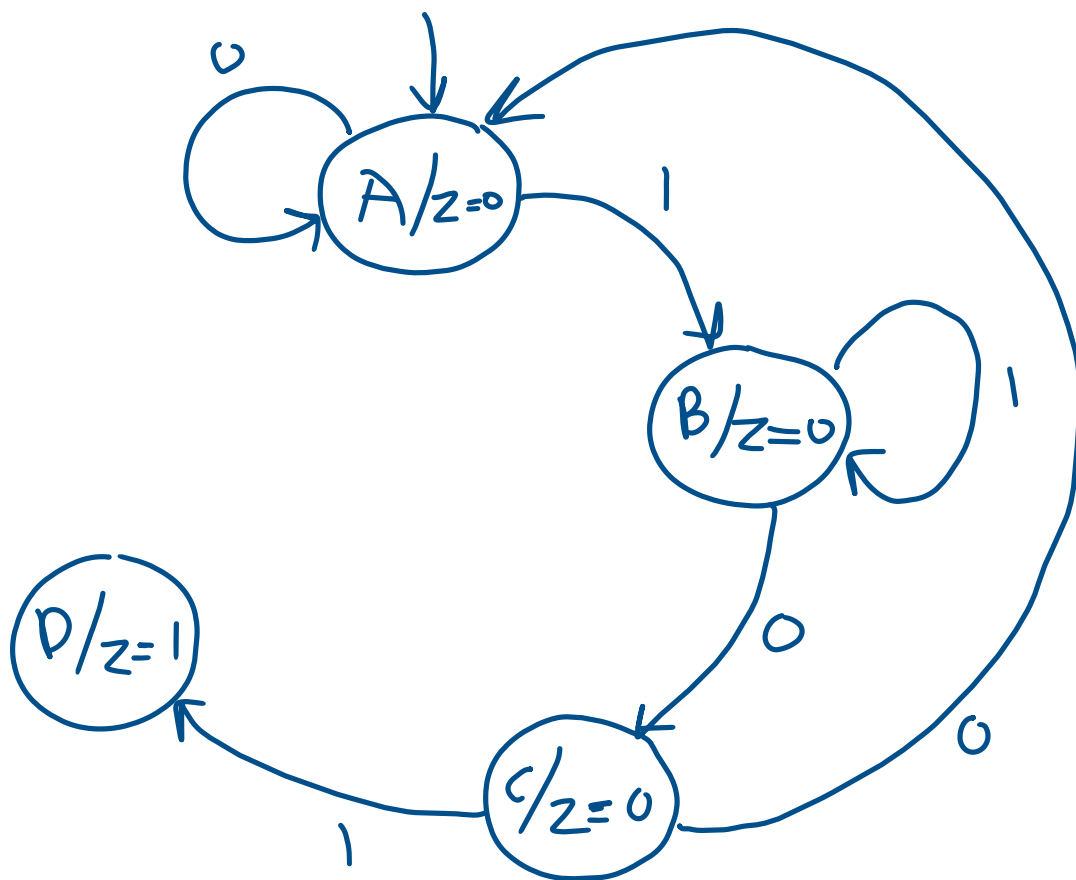
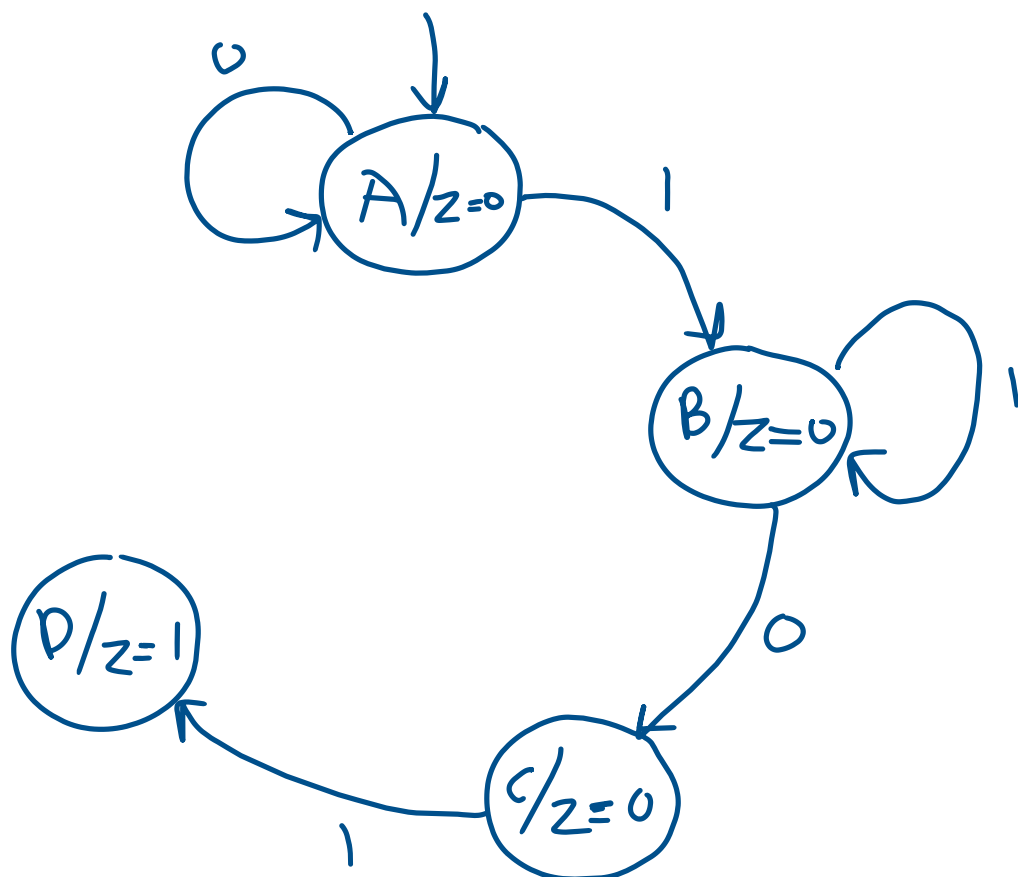
Overlapping is allowed, meaning that for each occurrence of 101, the output z becomes 1.

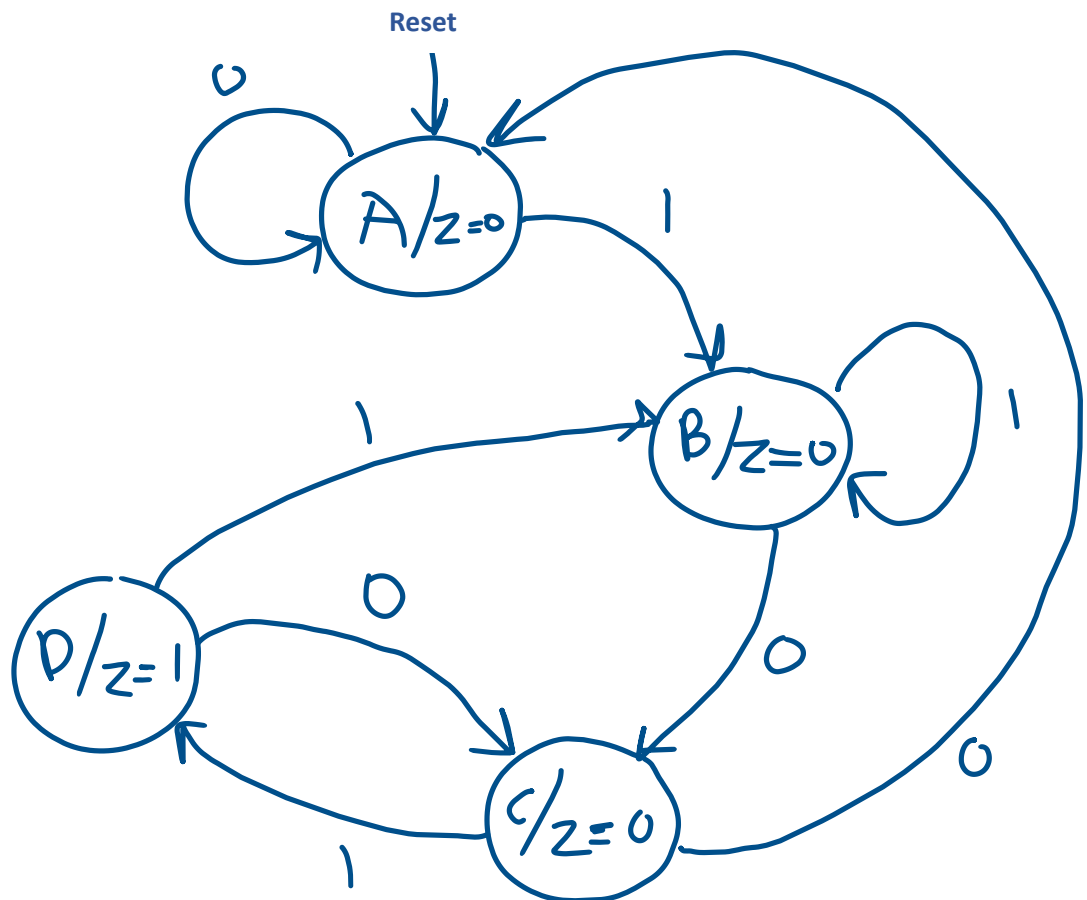
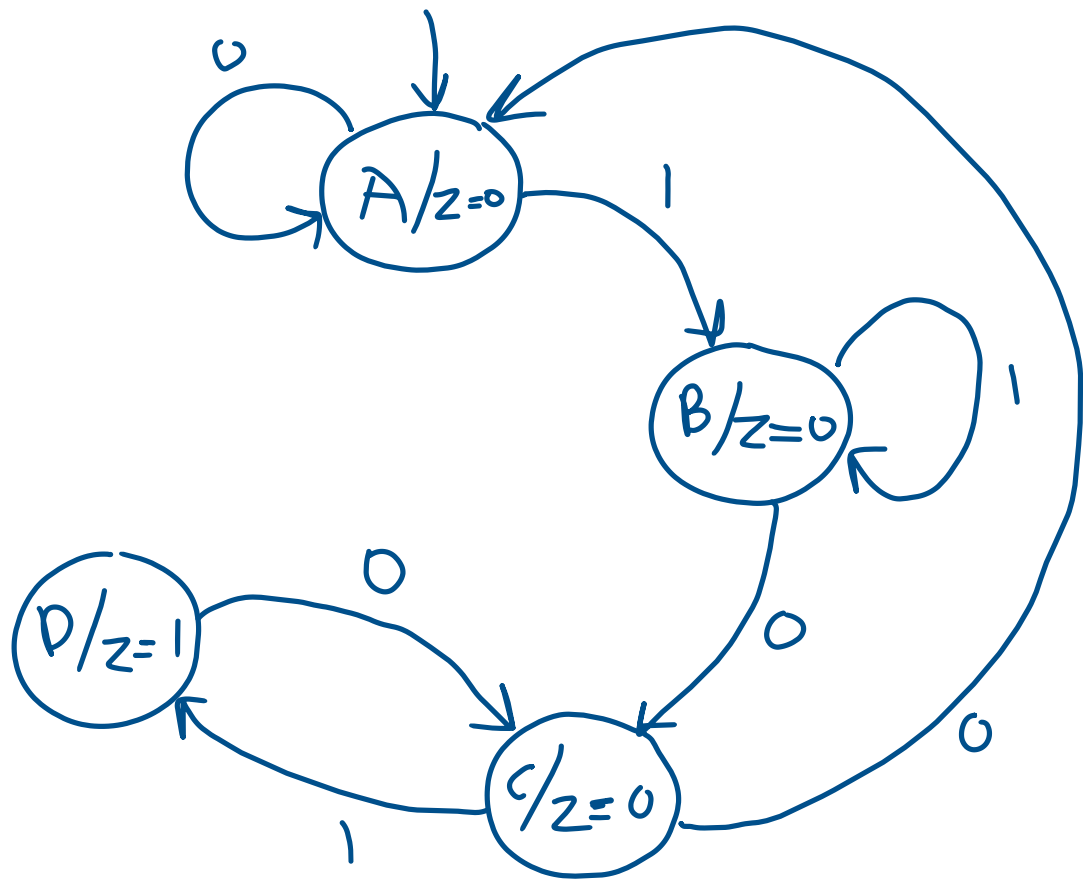
The above discussion is not part of the solution, but it clarifies the problem.

-State diagram (step by step):









-State table:

Present state	Next state		Output z
	W=0	W=1	
A	A	B	0
B	C	B	0
C	A	D	0
D	C	B	1

-State table assignment (I'll use standard encoding as a truth table)

4 states $\rightarrow$ 2 variables (2 flip flops)

Present state $y_2 y_1$	Next state		Output z
	W=0 $Y_2 Y_1$	W=1 $Y_2 Y_1$	
00	00	01	0
01	10	01	0
10	00	11	0
11	10	01	1

-Choose D flip-flop, therefore you do need a new assignment table. Fill out the K-maps.

$y_2 y_1$

w

	00	01	11	10
0	0	0	0	0
1	1	1	1	1

$$Y_1 = w$$

$y_2 y_1$

w

	00	01	11	10
0	0	1	1	0
1	0	0	0	1

$$Y_2 = \bar{w} y_1 + w y_2 \bar{y}_1$$

y_1

y_2

	0	1
0	0	0
1	0	1

$$z = y_2 y_1$$

-Draw the circuit:

